

CREATING HABITAT FOR KŌURA IN LAKES AND STREAMS: GUIDANCE FOR HABITAT ENHANCEMENT AND INFRASTRUCTURE DESIGN



REPORT PREPARED FOR CAWTHRON INSTITUTE

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Creating Habitat for Kōura in Lakes and Streams: Guidance for Habitat Enhancement and Infrastructure Design

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Cover image

Under calm conditions and with good water clarity, night-time spotlighting is a cost-effective method for monitoring kōura in the littoral zone of lakes and in small streams.

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EXECUTIVE SUMMARY

This report presents concept design ideas for incorporating freshwater crayfish (kōura) habitat features into lake and river flood protection, erosion control, and shoreline infrastructure. The aim is to demonstrate how ecological outcomes for kōura can be achieved through practical design approaches that can be incorporated into new infrastructure projects, retrofitted to existing assets, and integrated within wider restoration programmes.

Despite their ecological and cultural importance, kōura populations have declined in many lakes, rivers, and streams throughout Aotearoa New Zealand. These declines are largely attributed to human-induced pressures, including habitat loss and degradation, declining water quality, invasive predators, and changes to aquatic ecosystem structure and function.

Habitat creation and restoration projects have the potential to make a significant contribution to kōura recovery across Aotearoa. Many habitat enhancement measures are straightforward, cost-effective, and can be readily incorporated into infrastructure design and maintenance programmes. By considering kōura habitat requirements early in the design process, flood and erosion protection assets can provide both engineering and ecological benefits. Kōura require a combination of shelter, stable substrate, and access to food resources. During daylight hours, kōura typically occupy sheltered refuges and emerge at night to forage. The availability of refuge habitat is therefore one of the most important factors influencing kōura abundance and survival.

This report provides design principles and concept options for incorporating kōura habitat into:

1. Lakeshore walls and erosion protection structures.
2. Boat ramps, jetties, and associated recreational infrastructure.
3. River and stream flood protection and erosion control infrastructure.

The report also discusses opportunities to incorporate mātauranga Māori and kaitiakitanga, into infrastructure planning, design, construction, and monitoring programmes. Integrating these principles can help strengthen cultural outcomes, support biodiversity enhancement, and promote the long-term stewardship of freshwater ecosystems.

By combining engineering objectives with ecological and cultural values, flood and erosion protection infrastructure can contribute to the protection and recovery of kōura populations while enhancing the health and resilience of freshwater environments.

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1. INTRODUCTION

Aotearoa New Zealand (hereafter Aotearoa) has two species of freshwater crayfish commonly known as kōura. Their distributions are quite separate, with one found in the North Island and the West Coast of the South Island (*Paranephrops planifrons*), and the other in the east and south of the South Island (*P. zealandicus*). Kōura are an important component of freshwater ecosystems in Aotearoa and are recognised as a taonga species with significant ecological, cultural, and customary values. Kōura contribute to nutrient cycling, organic matter processing, and aquatic food webs, while also providing an important mahinga kai resource for tangata whenua and the wider community.

Despite their ecological and cultural importance, kōura populations have declined in many lakes and streams throughout Aotearoa, largely a result of human activities. Population declines have been associated with multiple stressors, including habitat loss and degradation, declining water quality, invasive predators, and changes in aquatic ecosystem structure. For example, in Lakes Rotorua and Rotoiti, located in the Bay of Plenty region, kōura populations have declined by 76% and 91%, respectively, over the past two decades, due to eutrophication, invasive macrophytes and predation by invasive catfish (*Ameiurus nebulosus*; Kusabs et al., 2026; Figure 1). Unless measures are taken to help the species, kōura range and abundance will continue to decline over the next few decades.

Habitat creation and site restoration projects have the potential to make a considerable contribution to kōura population recovery in Aotearoa. This can be straightforward and inexpensive, and can easily be incorporated into new infrastructure developments, retrofitted to existing infrastructure, and integrated into ecological restoration projects.

This document provides guidance on the principles of kōura habitat enhancement, including restoration approaches, creation of artificial refuge habitat, and opportunities to incorporate kōura habitat features into infrastructure design.



Figure 1 Brown bullhead catfish (*Ameiurus nebulosus*), an invasive predator of kōura (*Paranephrops planifrons*) in Lakes Rotorua and Rotoiti.

1.1 Kōura Biology and Ecology

Kōura are largely nocturnal, which helps them avoid predation by cormorants (shags) and fish such as rainbow trout. However, they are still vulnerable at night to nocturnal predators including tuna (eels) and, in shallow waters, terrestrial predators such as the introduced Norwegian rat (*Rattus norvegicus*). The crushed remains of kōura are commonly found in cormorant pellets and rat scats, and these can sometimes provide the first evidence of kōura presence in previously unsurveyed waters (Figure 2).

Kōura hide from predators in refuges on the bed or in the banks and use their prominent claws in defence if attacked. Juvenile kōura are particularly vulnerable to predatory fish but are also eaten by a range of predatory invertebrates, including other kōura.



Figure 2 Remains of three kōura following predation by a rat at Hinemoa's Point, Lake Rotorua, 14 April 2025.

Kōura have a varied, omnivorous diet, including leaf litter, submerged aquatic plants, other invertebrates (especially slow-moving ones such as aquatic snails, worms, some insect larvae and eggs) and dead fish. The actual diet depends on the age of the crayfish and the availability of different foods.

Kōura mate in autumn, females brood their eggs under their tails through the winter and spring, when crayfish are relatively inactive. Activity increases with warmer temperature in spring. After hatching, the young develop while still attached to the underside of the tail and are later released. Kōura are slow-growing and typically take three years to reach sexual maturity (at 21 mm orbital carapace length or more). Those that survive to maturity may live for ten years or more.

1.2 Understanding Kōura Habitat Requirements

Kōura require a combination of shelter, stable substrate, and access to food resources. During daylight hours, kōura typically occupy sheltered locations and emerge at night to forage. The availability of refuge habitat is therefore a critical factor influencing kōura abundance.

Important habitat features include:

- Shelter and refuge areas: Areas containing boulders, cobbles, gravel, woody debris, and submerged vegetation provide opportunities for shelter, feeding, and breeding.
- Complex shoreline structure: Natural shorelines with varied depths, substrates, and submerged vegetation provide greater habitat diversity than uniform banks or hardened edges.
- Stable substrates: Stable substrates such as large boulders and woody debris are critical in habitats which are exposed to large waves or high flows.
- Water quality: Good oxygen levels (> 5 mg/l), suitable temperatures (max. <25°C), and healthy aquatic ecosystems support kōura survival and reproduction.

Habitat enhancement should aim to recreate the complexity and diversity found in natural lake margins and streams but should also be present across a range of depths to account for water level variation.

2. Principles for creating kōura habitat

2.1 Increase Structural Complexity

The availability of suitable habitat is one of the few factors limiting kōura populations that can be readily improved. In many aquatic environments, kōura abundance is primarily determined by the availability of secure shelters (refuges), as food is rarely limiting due to their opportunistic diet. Kōura occupy a wide range of natural and artificial refuges, including rock crevices, woody debris, root masses, and even discarded items such as cans and bottles. Therefore, habitat enhancement can be an effective way to increase kōura populations by creating additional shelter opportunities that can reduce predation pressure. This can often be achieved using materials already available on site or by protecting and enhancing existing habitat. Habitat enhancement does not need to occur across an entire lake or stream; where resources or materials are limited, improving habitat within small bays or selected sections of shoreline may be sufficient to support healthy kōura populations. From the perspective of a kōura, criteria for a suitable refuge may be:

- Provide a big enough space to get the whole body inside, including claws, but not so roomy that a predator can gain access
- Stable enough to resist large waves or high flows
- Not liable to sudden loss of water or subject to large accumulations of aquatic weed/debris
- Easy to keep clean of excess silt, weed and debris
- Close to good foraging area

2.2 Protect and Enhance Existing Habitat

Existing habitat should be recognised and protected wherever possible. Retaining natural shoreline features, avoiding unnecessary disturbance, and maintaining riparian vegetation and submerged woody material can often provide greater ecological benefits than replacing with artificial habitat. Where natural habitat remains, enhancement can involve:

- Adding additional cover materials
- Protecting vulnerable areas from disturbance
- Reducing sediment inputs and erosion

2.3 Incorporating Habitat into Infrastructure Design

Infrastructure projects provide significant opportunities to incorporate kōura habitat at little additional cost. Traditional hard-engineered structures often create simplified environments, but small design modifications can greatly increase their ecological value. Before adding kōura habitat features, it is important to check with your local or regional to determine whether resource consent is required. Some councils already recognise the value of habitat enhancements and include provisions for kōura habitat additions in consent conditions. For example, the Bay of Plenty Regional Council provides guidance for lake structures that incorporate habitat features

<https://www.boprc.govt.nz/environment/resource-consents/applying-for-resource-consent/consent-forms-maps/lake-structure-consents/>).

Lakeshore walls and erosion protection

Vertical walls and rock revetments (riprap) can be modified to improve their function as kōura habitat by incorporating:

- Rock cavities and interstitial spaces that provide shelter
- A mixture of cobble and boulder sizes to create a range of refuge habitats (Figure 3)
- Rock or woody habitat that extend into deeper water where water levels fluctuate and shallow areas periodically dry out.

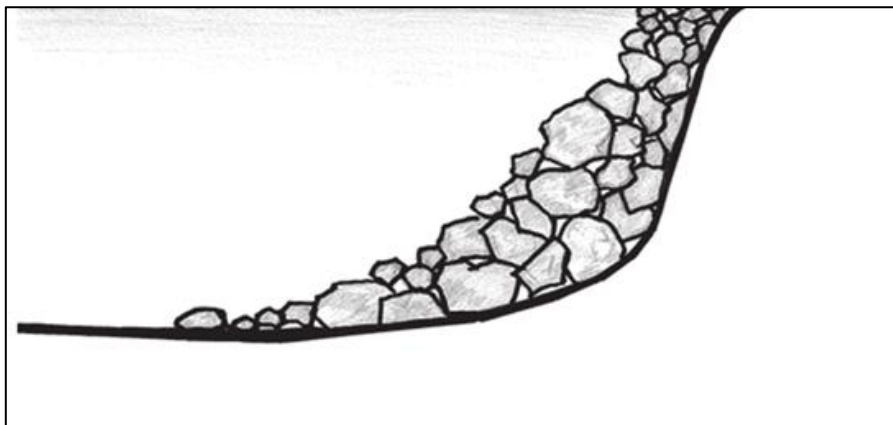


Figure 3 A mixture of large and small cobbles and boulders will create numerous refuges of different sizes for kōura at different life stages.

Boat ramps and jetties

Boat ramps and jetties provide good habitat for kōura. Kōura are often present under the concrete slabs constituting the boat ramps and are often found around the jetty piles.

Structures associated with lake access can incorporate:

- Rock and cobble habitat around jetty piles and concrete ramps and shoreline margins
- Wooden piles or poles with drilled holes to provide refuge habitat. These are particularly useful in marinas where boats are moored close to the shore and space for rock habitat is limited.
- Riparian planting along lakeshore margins to provide shading, bank stability, and organic inputs that support aquatic habitat.

Stream restoration and flood protection works

Stream engineering can incorporate kōura habitat by:

- Using a mixture of rock sizes in bank stabilisation structures (appendix 1a, 1b)
- Riparian plantings along the stream margins
- Avoiding excessive channel simplification
- Maintaining pools and slow-flow areas
- Adding large woody debris
- Creating undercut banks and refuge spaces.
- Adding wetted edge plantings within spaces between rocks within riprap walls (See: Holmes et al. 2026 for concept design)

Integrating Mātauranga Māori and Kaitiakitanga

Kōura are a taonga species, and habitat enhancement provides an opportunity to integrate mātauranga Māori alongside scientific approaches. Working with mana whenua can help identify:

- Important kōura habitats and historical knowledge
- Appropriate restoration priorities
- Preferred materials e.g., local rocks and untreated timber
- Cultural values associated with freshwater environments
- Monitoring approaches incorporating local knowledge and mātauranga
- Complementary monitoring methods such as whakaweku monitoring, tiro tiro (observation), and maramataka (Māori lunar calendar) to strengthen long-term stewardship.

Table 1 shows examples of habitat creation for kōura.

Table 1. Habitat creation and enhancement for kōura.

Refuge Type	Technique	Comments
Rocky Reefs	Lake reefs: Rock and cobbles piled into a mound (2.5m x 2.5m, 0.5m high) on the lake edge (>1m) or outside the weed beds on sand and stone substrate (Figure 4). Place large cobbles or boulders (>20 cm across) on the lakebed. Stack them to leave gaps that kōura and small fish can use as refuges.	Best in deeper areas that are not regularly exposed by fluctuating water levels. Use local stone where possible. A range of stone sizes (approximately 20–80 cm) provides a variety of refuge spaces. Can be logistically challenging, requiring specialist equipment e.g., a barge.
Boat Jetties	Jetty reefs – rocks and cobbles piled on top of sand/stone substrate beneath a boat jetty at depths <2m (Figure 4).	Best around non-public jetties as rocks can be interfered with around public/high use jetties. Use local stone where possible. A range of stone sizes (approximately 20–80 cm) provides a variety of refuge spaces. Cost effective as rocks can be deployed using a car trailer and wheelbarrow.
Lakeshore Habitat Walls	Place large boulders (>200 cm across) along the margins. Stabilise eroding slopes with large rocks (unconcreted). Use rough, unfaced stones so gaps remain large enough for kōura. (Figures 5 & 6).	Work best on banks where adjacent water is deeper than 30 cm. Can be expensive as requires specialist equipment to lay/ manoeuvre large boulders.
	Gabion baskets filled with small stones (15–25 cm) provide limited habitat for kōura (Figure 7). However, a rock apron constructed in front of the gabion baskets can create valuable habitat for both adult and juvenile kōura, especially when combined with riparian vegetation such as native shrubs, ferns, and Carex species. A suitable design is a rock apron approximately 2 m wide and 0.5m high, constructed from mixed stones 15–30 cm in diameter and placed along the base of the gabion baskets.	Leftover gabion stones can be used to create the rock <i>apron</i> in front of gabion baskets. This apron can be easily installed using a car trailer and wheelbarrow.
	Timber retaining walls provide no suitable refuge habitat for kōura. However, as with gabion baskets, habitat can be created by constructing a rock apron along the base of the wall and combining it with riparian planting. (Figure 7).	The most cost-effective type of habitat wall; however, it provides no suitable habitat for kōura. A rock apron constructed along the front of the timber wall is required to create refuges and provide suitable habitat for kōura.
Wood or vegetation along banks	Plant native trees, shrubs and grasses along selected sections of the bank to provide shade and for bank stabilisation (Figure 8).	Trees on slightly undercut vertical banks with large, submerged roots provide ideal habitat. Avoid excessive tree planting, as too much shade can reduce wetted edge and aquatic vegetation.
Artificial refuges	Where vertical banks require hard reinforcement, use engineering bricks (with holes), concrete blocks or poles with holes facing outward. Timber framed gabion baskets can provide kōura habitat while allowing for boat mooring and recreational use (Figure 9).	Wooden poles are particularly useful in marinas where boats are moored close to the bank.

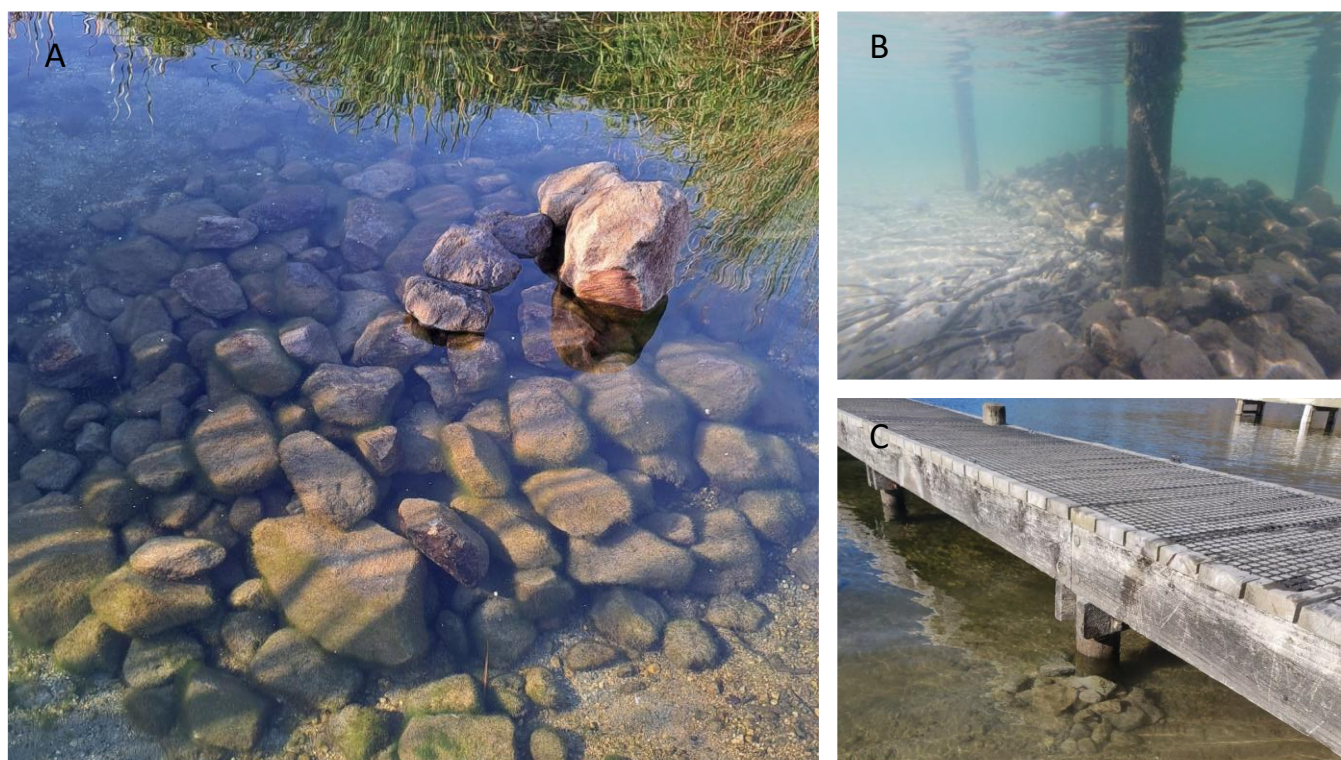


Figure 4 Examples of rocky reefs located (A) adjacent to and (B) beneath boat jetties. Rocks placed around (C) jetty piles and beneath a jetty create valuable kōura habitat while remaining compatible with recreational use and boat mooring.



Figure 5 Rock riprap provides excellent kōura habitat, particularly when constructed from a mix of cobbles and boulders and located adjacent to deeper water. (A) Lake Rotomā along State Highway 30, and (B) Ngongotaha Stream off Western Road.



Figure 6 Replacing a timber wall with a boulder habitat wall at Lake Rotoiti, note the submerged apron of cobbles in front of the boulders.



Figure 7 Timber (right) and gabion basket (left) walls provide limited habitat value for kōura and fish but can be enhanced by adding a submerged rock/stone apron at their base, creating additional shelter, cover, and habitat complexity.



Figure 8 Rock protection installed at a lake erosion site at Hamurana, Lake Rotorua. Although the shallow water depth limits its suitability as habitat for adult kōura, the rocks do provide refuge habitat for juvenile kōura and native fish species.



Figure 9 (A) Timber-framed gabion baskets filled with rock at Ōkawa Bay, Lake Rotoiti. When combined with (B) a submerged rock apron, these structures provide habitat for kōura and fish while maintaining access for boat mooring and recreational use.

3. Monitoring and evaluation

Monitoring is essential to determine whether habitat creation and management actions are achieving their intended objectives. Where practicable, a fisheries survey should be undertaken before any works commence to establish baseline conditions. Photographs of the site before, during, and after construction are also valuable for documenting changes over time. Monitoring the response of kōura and other aquatic species following habitat enhancement provides an opportunity to evaluate project success, improve understanding of habitat requirements, and refine the design of future restoration and infrastructure projects.

Monitoring should include:

- Presence and abundance of kōura
- Kōura size, sex, presence of eggs or juveniles and shell hardness (moulting)
- Use of created habitat features
- Any changes in surrounding habitat condition
- Presence and abundance of fish species

3.1 Monitoring methods

A range of methods can be used to monitor kōura populations within the littoral zones of lakes and in streams and rivers. The choice of method depends on site characteristics, water clarity, habitat complexity, accessibility, and available resources. Because kōura are nocturnal and occupy a variety of habitats, combining multiple sampling approaches can provide a more comprehensive assessment of population abundance, size structure, habitat use, and recruitment. Where possible, monitoring should also record the presence and relative abundance of fish species, particularly predatory or invasive species that may influence kōura populations. For reviews of kōura and fish monitoring methods see Kusabs et al. 2018 and Joy et al. 2013.

Spotlighting/Rama kōura

Night-time spotlighting is a simple and effective method for assessing the presence and relative abundance of kōura and fish in shallow littoral habitats and streams, particularly where water clarity is high (Figs. 10A & 11). Surveys are conducted after dark using a high-powered spotlight to scan suitable shoreline habitats, with observed kōura counted and, where possible, their size estimated and recorded. Under favourable conditions, kōura can also be captured using a dip net and bucket for measurement of OCL, assessment of sex and shell hardness (soft) and presence of eggs. While the effectiveness of spotlighting is influenced by water clarity and habitat complexity, it provides a rapid, non-invasive method for detecting kōura, assessing relative abundance, and identifying areas of high habitat use.

Minnow traps

Baited minnow traps can be used to capture kōura in lake shallows, as well as streams and rivers (Figure 10B). They are particularly useful where artificial shelters are unsuitable or where additional information on kōura population characteristics are required. Traps are typically baited and deployed

overnight before being retrieved and processed the following day. Captured kōura can be identified, measured, sexed, and released. Trap catches provide an estimate of relative abundance; however, catch rates can be influenced by bait type, trap design, soak duration, temperature, habitat characteristics, and the presence of competing or predatory species. Standardising trap deployment methods is therefore essential for comparisons among sites and over time.

Good practice includes using the same trap design throughout a survey programme (e.g., Gee-style minnow traps or collapsible bait traps), applying a consistent bait type and quantity (e.g., 20–50 g of trout flesh or cheese), deploying traps for a standard overnight soak period, and establishing fixed sampling sites that can be revisited over time. For long-term monitoring programmes, surveys are generally best conducted from spring through autumn, as kōura activity declines substantially during winter when water temperatures fall below approximately 10°C.

Fine mesh fyke nets

Fine mesh fyke nets provide an additional method for sampling kōura and are particularly useful for capturing larger fish species. Nets are typically baited and set overnight in shallow margins (Figure 10C). Captured kōura can be processed in the same manner as other methods, with measurements and biological information recorded before release. Fyke nets may also capture fish species, providing valuable information on the wider aquatic community and potential kōura predators. As with other minnow traps, net catches can be influenced by numerous factors; see ‘Minnow traps’ above. When using fyke nets, create air spaces for waterbirds that may inadvertently enter the nets by attaching buoys to the net chambers or raising the cod end above the water surface, reducing the risk of drowning (Figure 12).

Whakaweku (artificial kōura shelters)

Whakaweku, traditionally constructed from bundles of bracken fern (*Pteridium esculentum*), are artificial shelters designed to provide refuge habitat for kōura and are widely used for monitoring in both lakes and streams (Figure 10D). In lakes, multiple whakaweku are typically grouped together to form a tau kōura, which generally consists of 10 or more whakaweku. Tau kōura are used for harvesting and monitoring kōura in deeper habitats, at depths ranging from approximately 5 to 35 m.

Whakaweku are not particularly effective for monitoring kōura in shallow littoral lake habitats, where other methods such as night-time spotlighting, traps, and nets are more reliable. However, individual whakaweku are highly effective for monitoring kōura populations in streams and rivers. Whakaweku should be deployed for a minimum of two weeks before retrieval and inspection. The number of kōura occupying each whakaweku provides an index of relative abundance (CPUE). Captured kōura can also be measured, sexed, and assessed for reproductive condition and moulting status.

For a detailed description of the use of whakaweku for sampling kōura and fish in streams, refer to Kusabs et al. (2018) and <https://niwa.co.nz/freshwater/monitoring-koura/tau-koura-sample-collection-and-processing-protocol>.

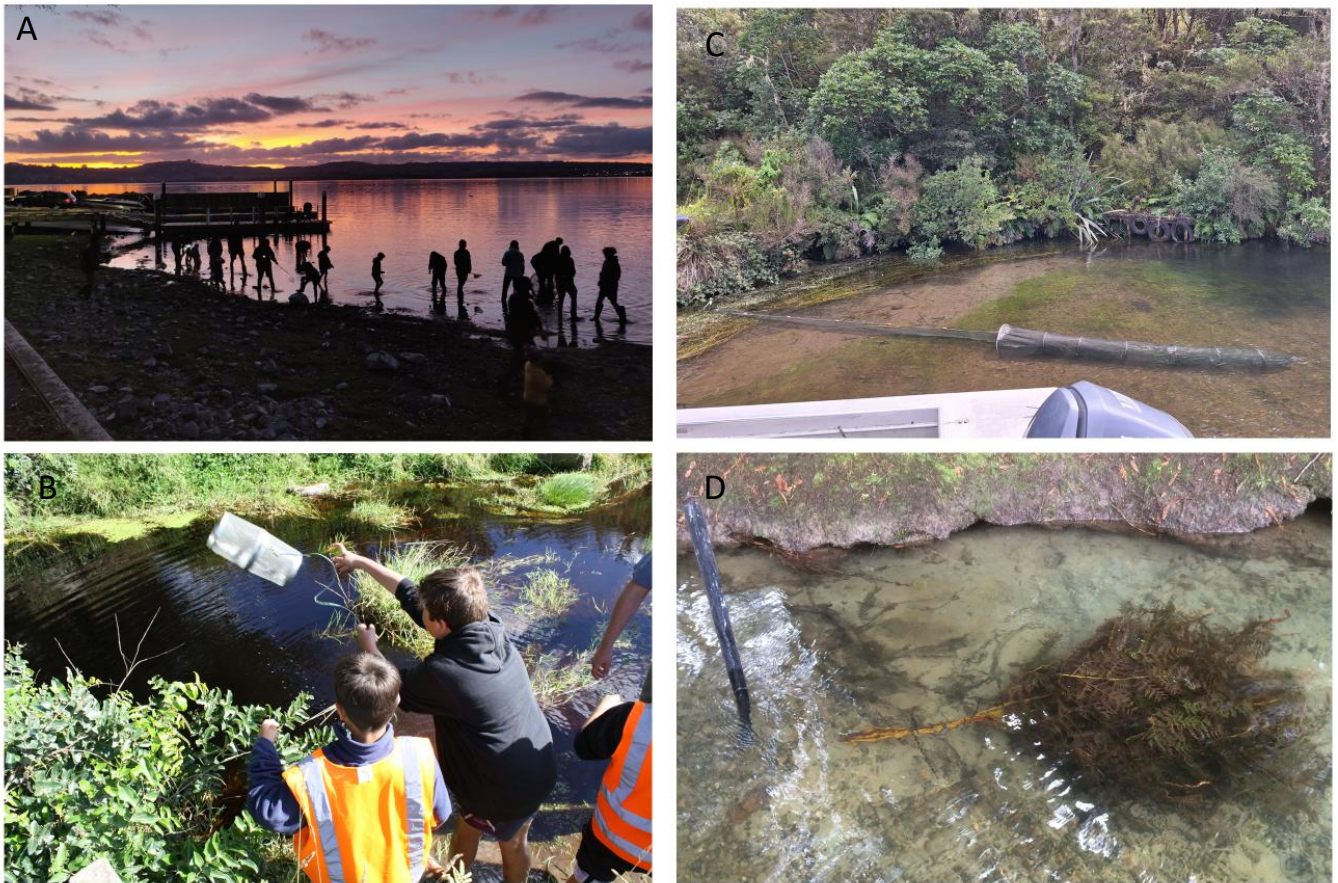


Figure 10 Four commonly used methods for surveying kōura (a) spot lighting, (B) minnow trapping, (C) fine mesh fyke netting and (D) whakaweku (bracken fern bundles).



Figure 11 Adult koaro (*Galaxias brevipinnis*) observed at night among boulder and cobble habitat at Lake Rotoiti. Koaro commonly occupy rocky littoral habitats in this lake and are frequently encountered during night-time spotlight surveys.

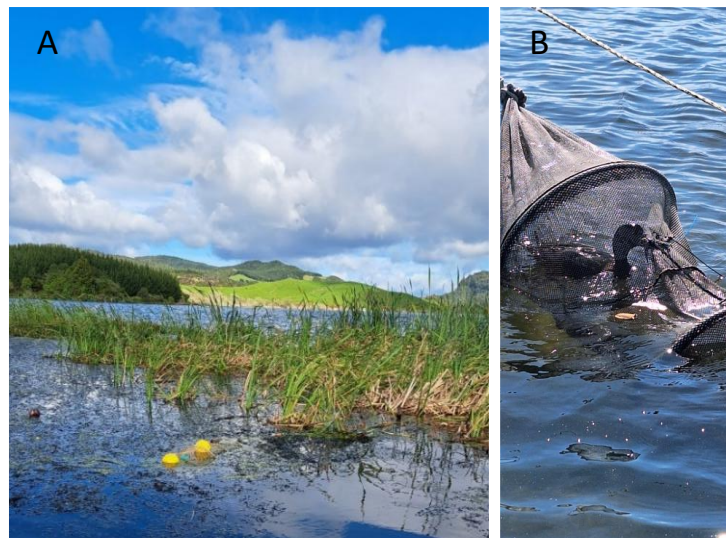


Figure 12 When using fyke nets, create air spaces for waterbirds that may inadvertently enter the nets by (A) attaching buoys to the net chambers or raising the cod end above the water surface, reducing the risk of drowning. (B) A scaup accidentally captured in a fine mesh fyke net at the Lake Rotorua lakefront.

3.2 Measuring and sexing kōura

1. Measure kōura - the orbital carapace length (OCL) is measured from behind the eye to the end of the carapace (above the tail) along the top and centre of the back (Figure 13A).
2. Determine sex of kōura, and presence of eggs/hatchlings on females (see Figure 13B).
3. The following power regression equations can be used to estimate kōura wet weight:

For **lake kōura** $W (g) = 0.0005 OCL^{3.0765}$ (Kusabs et al 2026), and

For **stream kōura** $W (g) = 0.000648 OCL^{3.0743}$ (Riordan 2000).

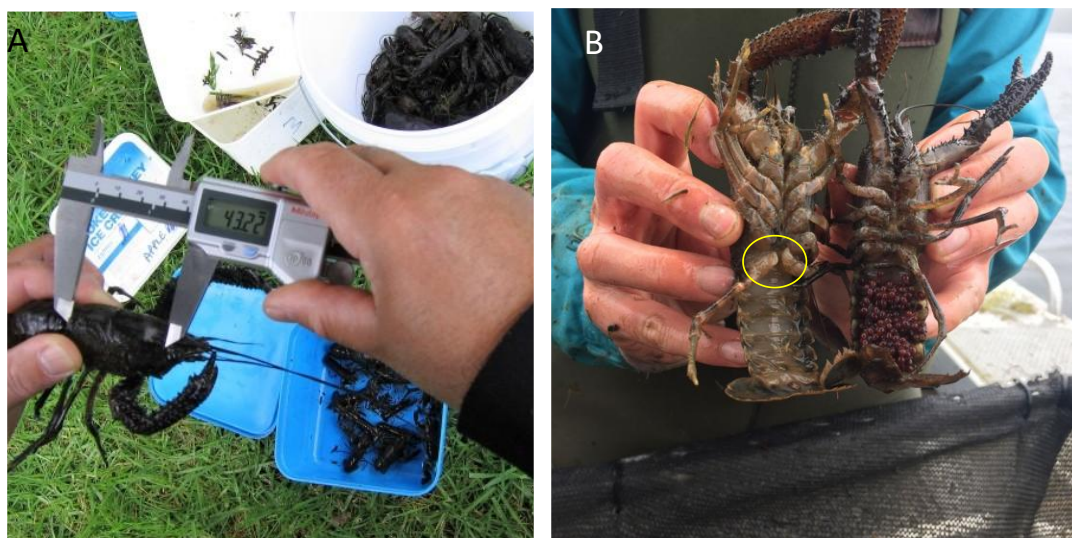


Figure 13 Measurement of orbital carapace length (A) and identification of male and female kōura (B) Note the gonopores/lumps at the base of the male's hind legs (yellow circle), the presence or absence of these features can be used to identify sex.

4. Summary

Thoughtful incorporation of ecological design can move infrastructure projects beyond minimising impacts to actively enhancing kōura habitat (Appendix 1). By recognising the importance of complex littoral and stream habitats, protecting existing habitats and incorporating kōura-friendly features, we can increase habitat availability and support resilient kōura populations.

Monitoring is essential to determine whether habitat enhancement projects achieve their intended outcomes. Monitoring should evaluate both habitat development and biological responses. Regular monitoring helps identify which habitat features and management practices are effective and where improvements may be needed. Each habitat enhancement project provides an opportunity to refine techniques, improve future designs, and share best practice.

The success of habitat creation is ultimately measured by the establishment of a healthy kōura population at an abundance appropriate to the amount and quality of available habitat. However, population establishment takes time. Kōura may be difficult to detect during the first few years after habitat creation, with populations typically becoming evident after around three to five years. It may take considerably longer for habitats to reach their full carrying capacity and support a stable, self-sustaining population.

5. Recommendations

The following principles are recommended for kōura habitat enhancement projects:

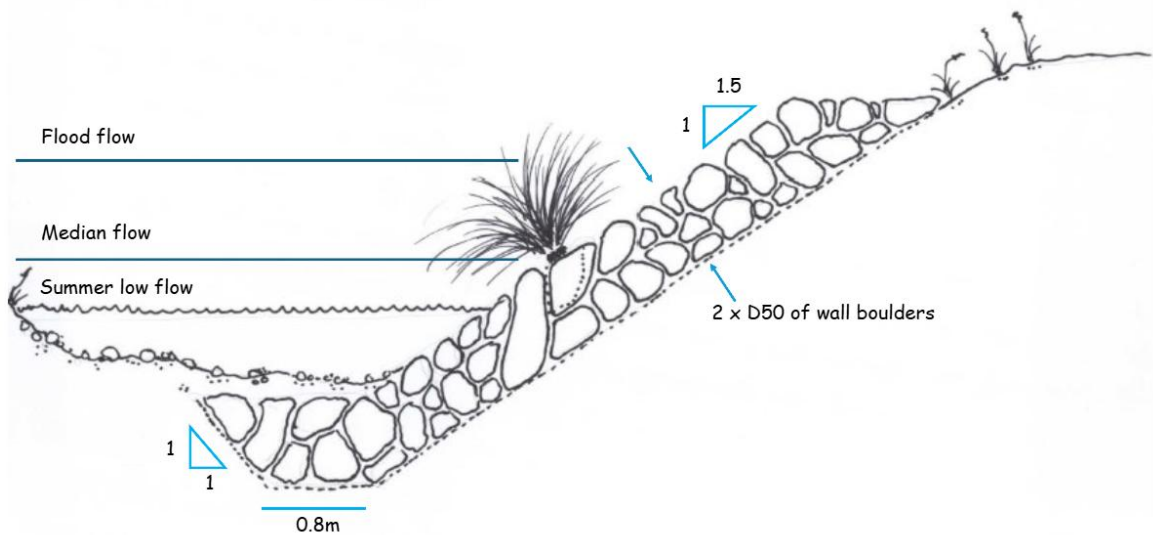
1. **Protect existing habitat** – Prioritise the protection and retention of existing kōura habitat before creating new habitat.
2. **Integrate habitat into project planning** – Consider kōura habitat requirements during the earliest stages of infrastructure planning and design.
3. **Enhance structural complexity** – Increase habitat complexity by incorporating natural materials or appropriately designed artificial refuge structures that provide shelter for a range of kōura life stages.
4. **Use suitable materials** – Use locally appropriate materials and avoid structures that simplify habitat.
5. **Partner with mana whenua** – Incorporate mātauranga Māori and work collaboratively with mana whenua throughout project planning, implementation, and long-term stewardship.
6. **Adhere to regional authority guidelines** – Confirm with your council/unitary authority if existing consent guidelines are in place for in-lake or in-stream structures.
7. **Minimise construction impacts** – Undertake pre-construction surveys to identify kōura populations and, where necessary, safely relocate kōura from work areas before physical works commence (<3 days).
8. **Monitor and adapt** – Monitor ecological outcomes and use results to improve future designs.

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Appendix

Appendix 1a. Concept design for interplanted stream rock wall to support aquatic values, see Holmes et al 2025 for more details



Appendix 1b. Idealised cross-section of stream showing the relative size range of different sized rocks and their vertical placement within a stream or lake bank to support multiple aquatic values.

